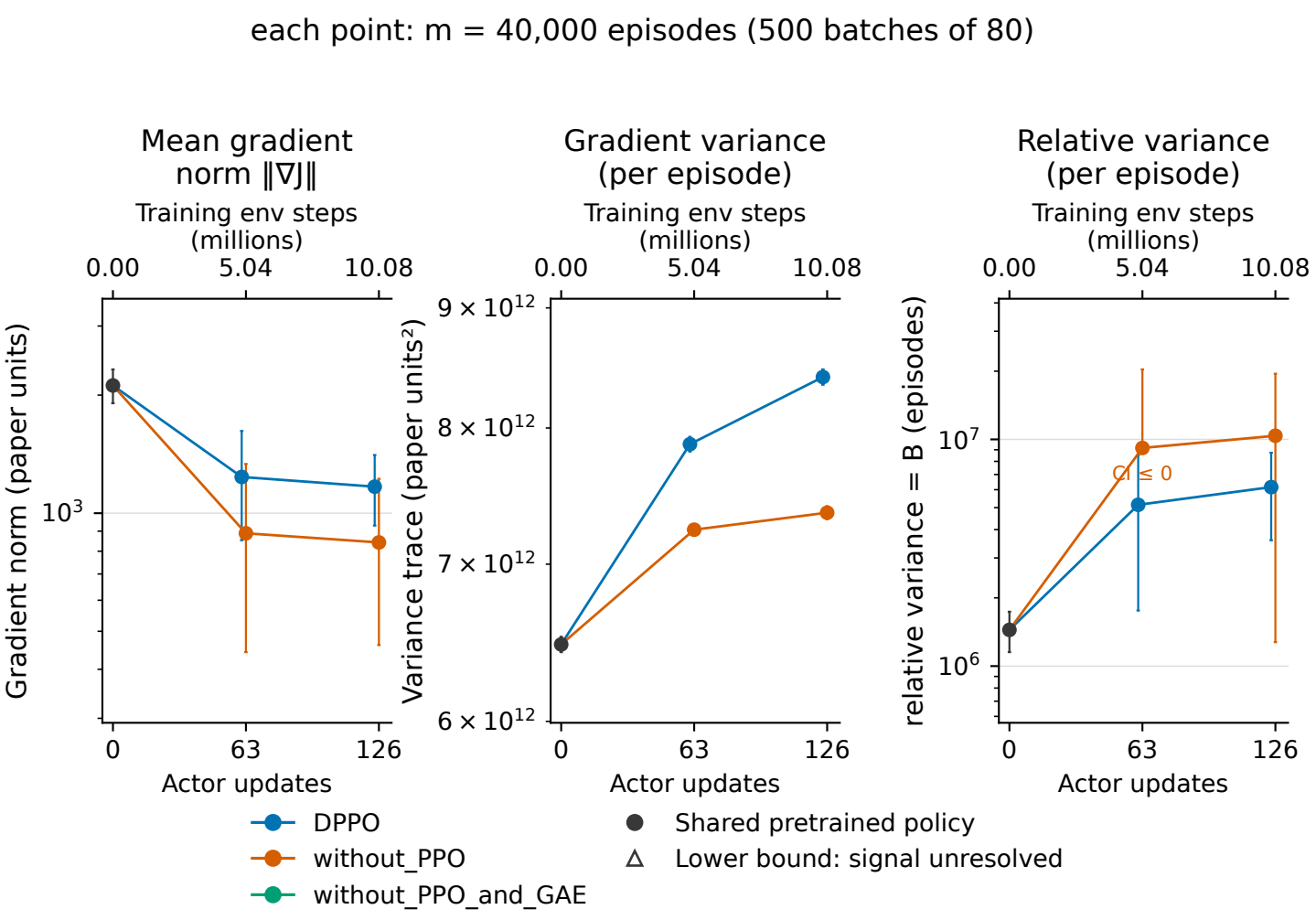
each point: m = 40,000 episodes (500 batches of 80)
Mean gradient norm ‖∇J‖
Training env steps (millions)
0.00
5.04
10.08
Gradient norm (paper units)
10^3
0
63
126
Actor updates
Gradient variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
Variance trace (paper units²)
9×10^{12}
8×10^{12}
7×10^{12}
6×10^{12}
0
63
126
Actor updates
Relative variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
relative variance = B (episodes)
10^7
10^6
CI ≤ 0
0
63
126
Actor updates
DPPO
without_PPO
without_PPO_and_GAE
Shared pretrained policy
Lower bound: signal unresolved